PRACTICAL WORKBOOK COMPANION

LibreOffice Calc Field Manual

Beginner-to-advanced practical spreadsheet guide

COVERS

- Excel-to-Calc transition
- Formulas
- Graphs
- Troubleshooting
- Data preparation
- Analysis
- Documentation

THE WORKFLOW IN ONE LINE

IMPORT | INSPECT | CLEAN | FILTER | CALCULATE | ANALYZE | GRAPH | VALIDATE | SAVE

Targets modern LibreOffice Calc. Menu paths can differ slightly between releases and systems.

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Purpose

This manual is for someone who knows some Microsoft Excel but must work in LibreOffice Calc. It is written as a practical workbook companion: follow the steps, copy the formulas, test them on a small sample, then apply them to the full dataset.

The examples use neutral measurement, inventory, operations, and financial-style tables. No specialist subject knowledge is required.

> **Version note** LibreOffice changes over time. This guide targets modern Calc, with menu paths based on the current menu interface. Some commands move slightly between releases, operating systems, and user-interface layouts. If a menu path differs, use **Help > Search Commands**, or press `Alt+?` if your installation provides command search. Press `F1` for the built-in Help. The function wizard is available with `Ctrl+F2`.

1 The Mental Model: Excel versus Calc

The important point: most spreadsheet concepts are identical. The differences are mainly interface layout, formula separators, sheet-reference notation, some function names, and compatibility details.

Excel concept	Calc equivalent / difference
Ribbon	Menus and toolbars by default; optional Notebookbar layouts exist.
Worksheet	Sheet
Workbook	Spreadsheet document
`.xlsx`	Opens and saves in Calc; use ODS as the native format.
`.xlsm`	May open, but VBA macros are not guaranteed to work.
`.ods`	Calc's native OpenDocument spreadsheet format.
Formula argument comma	Usually semicolon in Calc's default formula syntax.
`Sheet1!A1`	Usually `Sheet1.A1` in Calc's native syntax.
PivotTable	Pivot Table (historically called DataPilot).
Power Query	No direct built-in equivalent; use Calc tools, Python, database or ETL tools.
VBA	LibreOffice Basic / UNO API / supported scripting languages.
Chart Design ribbon	Double-click chart; use menus and right-click context commands.
Name Manager	Sheet > Named Ranges and Expressions > Manage (menu wording varies).
Goal Seek	Tools > Goal Seek.
Scenario Manager	Tools > Scenarios.

1.1 The interface

Calc normally has:

- **Menu bar:** File, Edit, View, Insert, Format, Styles, Sheet, Data, Tools, Window, Help.
- **Standard toolbar:** save, undo, redo, print, and common actions.
- **Formatting toolbar:** font, size, alignment, borders, number formats.
- **Formula bar:** name box, function wizard, input line.
- **Sheet tabs** at the bottom.
- **Status bar** at the bottom.

Excel users often search for a ribbon button. In Calc, learn the menu names:

Data	Sorting, filters, Pivot Tables, statistics, validity.
Insert	Charts, functions, images, rows/columns.
Format	Cells, rows, columns, conditional formatting.
Sheet	Sheet operations and named ranges.
Tools	Goal Seek, Scenarios, macros, options.

1.2 File formats and safe compatibility

Use ODS as your working master file:

```
File > Save As > ODF Spreadsheet (.ods)
```

When another person requires Excel:

```
File > Save As > Excel 2007–365 (.xlsx)
```

BEST PRACTICE

1. Keep one untouched source CSV or XLSX.
2. Save a working copy as ODS.
3. Do the analysis in ODS.
4. Save an XLSX delivery copy only when needed.
5. Reopen the XLSX copy and inspect formulas, charts, conditional formatting, and print layout.

Do not assume that every Excel feature survives conversion. Complex Excel macros, Power Query queries, external data connections, specialized charts, and some advanced formula behavior may change. If the file must return to Excel, test the round trip early.

1.3 Performance habits

For large tables:

- Avoid whole-column array calculations repeated thousands of times if a bounded range is enough.
- Prefer one clean source table with one header row and no blank rows inside it.
- Avoid merged cells in data tables.
- Avoid volatile formulas such as `NOW()` and `INDIRECT()` when not needed.
- Do not use repeated formatting across a million empty rows.
- Use helper columns for complex logic instead of one enormous nested formula.

- Turn off automatic recalculation temporarily only when a workbook is genuinely slow: use **Tools > Cell Contents > AutoCalculate** if present, or the calculation options under **Tools > Options > LibreOffice Calc > Calculate** (options vary by version). Restore automatic calculation before final checks.
- Save frequently.
- Use CSV for raw data exchange and ODS for the working analysis.

2 Set Up Calc Before Doing Serious Work

2.1 Confirm your version

```
Help > About LibreOffice
```

Record the version number. Function availability and menu names can depend on the version.

2.2 Set formula syntax and separators

```
Tools > Options > LibreOffice Calc > Formula
```

This area controls formula syntax and separators. In many default installations:

- Function arguments use semicolons: `=IF(A1>0;"Yes";"No")`
- Array-column and array-row separators have their own settings.
- Excel A1 syntax may be selectable.

If a formula copied from a website uses commas, do not assume the formula itself is wrong. Replace argument-separator commas with semicolons if your Calc uses semicolons. Do not replace commas inside quoted text.

2.3 Set calculation behavior

```
Tools > Options > LibreOffice Calc > Calculate
```

This is where you can inspect settings such as:

- Automatic calculation.
- Wildcards versus regular expressions in formulas.
- Iterative references, if required.
- Precision and calculation options.

2.4 Useful first settings

- **View > Freeze Rows and Columns:** keep headers visible.
- **View > Show Formula:** temporarily display formulas instead of results.
- **Tools > Options > LibreOffice > General:** extended tips can help identify toolbar buttons.
- **Tools > Options > LibreOffice > View:** icon style and scaling options.

3 Cells, References, Formulas, and Separators

3.1 Cell addresses

`A1`	Column A, row 1.
`A1:A10`	A range from A1 through A10.
`A:A`	The whole column A.
`$A$1`	An absolute reference.
`$A1`	Locks the column.
`A$1`	Locks the row.

Example: in C2, `=A2*B2`. Copying down changes it to `=A3*B3`.

Absolute-reference example: if the tax rate is in F1, use `=E2*$F$1` and copy down. F1 remains fixed.

> **F4** While editing or selecting a cell reference in a formula, press **F4** to cycle through relative, absolute, and mixed references in common Calc configurations. If it does not, insert dollar signs manually.

3.2 Sheet references

In native Calc syntax:

```
=Data.A2
=Data.$A$2
='Clean Data'.A2
```

Excel commonly displays `=Data!A2`.

If you open an Excel workbook, Calc may preserve or display a compatibility syntax depending on settings. Do not panic if the punctuation differs.

3.3 Formula anatomy

Every formula begins with `=`.

Operator	Meaning	Operator	Meaning
`+`	add	`=`	equal
`-`	subtract	`<>`	not equal
`*`	multiply	`>`	greater than
`/`	divide	`<`	less than
`^`	power	`>=`	greater than or equal
`&`	join text	`<=`	less than or equal

Example: `=(B2*C2)*(1-D2)`

Order of operations follows standard arithmetic. Use parentheses when in doubt.

3.4 Argument separators

Typical Calc	Typical Excel
`=SUM(A1;A10)`	`=SUM(A1,A10)`
`=IF(A1>10;"High";"Low")`	`=IF(A1>10,"High","Low")`
`=SUMIFS(C2:C100;A2:A100;"North";B2:B100;">100")`	`=SUMIFS(C2:C100,A2:A100,"North",B2:B100,">100")`

The semicolon is usually the only visible syntax difference in simple formulas, but not the only compatibility difference between the applications.

3.5 Formula Wizard

Press **Ctrl+F2** or click the Function Wizard button beside the formula line. Use it when:

- You do not know the exact argument order.
- You need to inspect required versus optional arguments.
- You want to browse categories.
- A formula returns an error and you suspect an argument is missing.

4 Import CSV Files Correctly

> **Why it matters** This is one of the most important practical skills. A wrong import setting can corrupt dates, decimal values, IDs, or columns before analysis even begins.

4.1 Import procedure

1. Open Calc.
2. **File > Open.**
3. Select the `.csv` file.
4. The Text Import dialog should appear.
5. Choose the correct separator: comma, semicolon, tab, or another delimiter.
6. Inspect the preview. Each field should appear in the correct column.
7. Set text encoding, usually UTF-8 unless the source says otherwise.
8. Check quoted text handling.
9. Set column types where necessary.
10. Click OK.

4.2 Important import choices

Separated by	Choose the actual delimiter.
Detect special numbers	Useful for dates and numbers, but inspect the result.
Quoted field as text	Important when text contains commas.
Language / locale	Controls decimal and date interpretation.
Column type: Text	Use for IDs with leading zeros, codes, or values that must not be converted.

Example: an ID `00127` must remain `00127`. Import that column as Text. Otherwise Calc may convert it to `127`.

4.3 Decimal separator problems

If the source uses `12.45` but your locale expects `12,45`, values may import as text or be misread. Fix:

- Reimport with the correct language/locale, or
- Select the column and use **Data > Text to Columns**, setting the correct separators and column type.

Do not blindly replace every period or comma; first identify which character is the decimal separator and which is the thousands separator.

4.4 If all data appears in one column

Cause: wrong delimiter. **Fix:**

1. Select the column.
2. **Data > Text to Columns.**
3. Choose the correct separator.
4. Inspect the preview.
5. Confirm.

4.5 If dates are imported as text

- Reimport and select a matching date format if possible.
- Or use **Data > Text to Columns** and select the date column type.
- If text is consistently `YYYY-MM-DD`, `DATEVALUE` may work depending on locale.
- Verify with `=ISNUMBER(A2)`. A real date is stored as a number; text dates are not.

5 Structure a Dataset So Formulas and Charts Work

Use a clean rectangular table.

GOOD STRUCTURE

Record_ID	Date	Group	Category	X_Value	Y_Value	Uncertainty
R001	…	North	A	12.4	8.1	0.2
R002	…	South	B	13.1	7.9	0.3

RULES

- One header row.
- One record per row.
- One variable per column.
- No merged cells inside the data.
- No blank rows inside the table.
- No subtotal rows inside raw data.
- Consistent units in each column.
- Do not mix numbers and text in the same measurement column.
- Keep raw source data separate from cleaned data.
- Add calculated columns to the right of the raw fields.

- Include units in header names where useful.

5.1 Convert a range into a manageable database range

Select the dataset, then **Data > Define Range** (wording may vary). Give it a name. This can help with sorting, filtering, and formulas.

5.2 Freeze the header row

Click cell A2 (or the row below the headers), then **View > Freeze Rows and Columns**. Depending on the interface, choose Freeze First Row or freeze rows/columns at the current cursor position. Test by scrolling.

5.3 Useful formatting

Format > Cells (`Ctrl+1`)

- **Numbers:** decimal places, dates, percentages, currency.
- **Font:** readable type and size.
- **Borders:** use lightly.
- **Alignment:** set headers and values consistently.

> **Note** Formatting changes appearance, not the underlying value. A value displayed as `0.05` may contain more decimal places internally.

6 Data Cleaning: Text, Numbers, Dates, Blanks, Duplicates

> **Rule one** Use a copy of the raw data. Never destroy the only original.

6.1 TRIM: remove extra spaces

```
=TRIM(A2)
```

Example: `" North Region "` becomes `"North Region"`.

TRIM removes leading/trailing spaces and reduces repeated ordinary spaces. It may not remove non-breaking spaces copied from websites. For those, use SUBSTITUTE with `CHAR(160)`:

```
=TRIM(SUBSTITUTE(A2;CHAR(160);" "))
```

6.2 CLEAN: remove non-printing characters

```
=CLEAN(A2)
```

Useful when text contains hidden control characters from imported files. A robust cleanup may be:

```
=TRIM(CLEAN(A2))
```

6.3 CONCATENATE and &

Join text:

```
=CONCATENATE(A2;" ";B2)
=A2&" "&B2
```

Modern versions may also support `CONCAT` or `TEXTJOIN`. If a function is unavailable in your version, use `&` or `CONCATENATE`.

6.4 LEFT, RIGHT, MID, LEN

Formula	What it does
`=LEFT(A2;3)`	First 3 characters
`=RIGHT(A2;4)`	Last 4 characters
`=MID(A2;2;5)`	5 characters starting at position 2
`=LEN(A2)`	Number of characters

6.5 FIND and SEARCH

Formula	What it does
`=FIND("-";A2)`	Case-sensitive position
`=SEARCH("ab";A2)`	Case-insensitive position

6.6 SUBSTITUTE versus REPLACE

SUBSTITUTE changes matching text:

```
=SUBSTITUTE(A2;".";"-")
```

Example: `12.03.2026` becomes `12-03-2026`.

REPLACE changes characters by position:

```
=REPLACE(A2;4;2;"XX")
```

Meaning: in A2, start at character 4, replace 2 characters with `XX`.

> **Do not confuse** **SUBSTITUTE** = replace matching text. **REPLACE** = replace a position-based segment.

6.7 UPPER, LOWER, PROPER

```
=UPPER(A2)
=LOWER(A2)
=PROPER(A2)
```

Useful for standardizing inconsistent names or category labels. Check abbreviations after PROPER; it may capitalize them in an unwanted way.

6.8 REGEX

Calc supports REGEX in modern releases. Availability and syntax can depend on version. General syntax:

```
=REGEX(Text; Regular_Expression; Replacement; [Flags])
```

Example	Purpose
`=REGEX(A2;"[^0-9]";"";"g")`	Remove all characters except digits
`=REGEX(A2;"[ ]+";" ";"g")`	Replace repeated spaces with one space
`=REGEX(A2;"^TMP-";"")`	Replace (remove) a leading prefix

Extract or replace patterns only after testing the expression on a few rows. The `"g"` flag means global replacement. In older versions without REGEX, use SUBSTITUTE, FIND, LEFT, MID, RIGHT, and helper columns.

> **Regex settings** **Tools > Options > LibreOffice Calc > Calculate.** Look for the regular-expression/wildcard settings. The setting affects how criteria in functions such as COUNTIF and SUMIF interpret pattern characters. The REGEX function itself is a separate function.

6.9 Remove punctuation or special characters

Example: preserve letters, digits, spaces, and hyphens:

```
=REGEX(A2;"[^A-Za-z0-9 -]";"";"g")
```

If the data contains accented or non-English letters, a simple A-Z pattern may remove valid characters. Test carefully and use a Unicode-aware pattern if supported by the installed regex engine.

6.10 Remove duplicates

Work on a copy or ensure the original is backed up.

COMMON MODERN ROUTE

1. Select the complete data range, including headers.
2. **Data > More Filters > Remove Duplicates** (if present in your version).
3. Choose which columns define a duplicate.
4. Confirm.

FALLBACK ROUTE (VERSIONS WITHOUT A DIRECT REMOVE DUPLICATES COMMAND)

1. Select the range.
2. **Data > More Filters > Standard Filter.**
3. Open Options.
4. Enable **No duplication**.
5. Choose where the filtered result should be copied if that option is offered.
6. Apply the filter.
7. Inspect the result before replacing the source.

ALTERNATIVE HELPER-COLUMN METHOD

```
=COUNTIF($A$2:A2;A2)
```

Rows where the result is greater than 1 are later occurrences of that ID. Filter the helper column to values greater than 1, inspect, then remove only confirmed duplicates.

> **Important** A duplicate is not always an error. Two separate transactions may share the same customer ID. Define the unique key first (for example, transaction ID, or a combination of date + reference + amount).

6.11 Missing values

First distinguish these – they are **not** equivalent:

- Truly empty cell.
- Empty string returned by a formula.
- A space character.
- Text such as `NA`, `N/A`, `NULL`, or `-`.
- Numeric zero.

FIND BLANKS

- Select the target column.
- **Edit > Find** (`Ctrl+F`) or **Edit > Find and Replace** (`Ctrl+H`).
- Use regular-expression mode and `^$` to search for empty cells, if supported by your search settings.
- Alternatively use a helper formula.

Formula	What it does
`=IF(A2="";"MISSING";"OK")`	Flag missing values
`=IF(A2="";"Unknown";A2)`	Replace missing text with a label
`=COUNTBLANK(A2:A1000)`	Count blanks
`=COUNTA(A2:A1000)`	Count non-empty cells

Do not replace missing measurements with zero unless zero is genuinely the correct value. Better choices may be:

- Leave blank.
- Use "Missing".
- Use a documented imputation method.
- Exclude the record with a clear reason.

6.12 Fill values down

To copy the value/formula from the top cell through selected cells:

1. Put the value/formula in the first cell.
2. Select that cell and the cells below it.
3. **Sheet > Fill Cells > Down**, or use `Ctrl+D` in common Calc configurations.

For a formula, relative references adjust row by row. To fill right, use **Sheet > Fill Cells > Right**. Keyboard shortcut can vary by configuration.

6.13 Convert text numbers to numbers

Symptoms: SUM ignores values, numbers align left, or ISNUMBER returns FALSE. **Options:**

- **Data > Text to Columns**, then set column type Standard.
- Use `VALUE(A2)` where appropriate.
- Multiply by 1 in a helper column: `=A2*1`, only if A2 is convertible.
- Check decimal separators and locale first.

6.14 Find and Replace

`Ctrl+H` opens Find and Replace. Use it to standardize:

- "NORTH" to "North"
- "." to "-"
- repeated labels
- known unwanted prefixes

Use Replace All only after checking the scope and previewing a few examples. Regex mode changes the meaning of special characters.

7 Filtering, Sorting, Selecting, and Filling

7.1 AutoFilter

1. Click in the data table.
2. **Data > AutoFilter.**
3. Use the dropdown arrows in the header row.
4. Select values or conditions.
5. To remove filters, use Data > AutoFilter again or clear the filter.

7.2 Standard Filter

Data > More Filters > Standard Filter. Use it for multiple conditions such as:

- Region equals North AND value greater than 100.
- Category equals A OR category equals B, depending on how criteria rows are arranged.

7.3 Sorting

1. Select the **entire table**, not just one column.
2. **Data > Sort.**
3. Choose the sort key.
4. Add secondary sort keys if needed.
5. Confirm that the range has headers.
6. Sort.

> **Warning** Sorting only one column can detach values from their records. Select the whole table.

7.4 Select visible cells only

When copying filtered data, hidden rows can sometimes be included depending on the command. Use **Edit > Select Special > Visible cells only** if available in your version, or consult Help for "Only Copy Visible Cells".

7.5 Conditional formatting

Format > Conditional > Condition (menu wording varies). Example uses:

- Highlight values above a threshold.
- Color negative values.
- Use data bars or color scales if available.

Conditional formatting is visual; it does not alter the value.

8 Formula Toolkit: The Functions You Are Likely to Need

> **Note** Formula syntax in this manual uses semicolons. If your Calc is configured differently, adapt the separator.

8.1 Basic arithmetic

`=A2+B2`	`=A2-B2`	`=A2*B2`	`=A2/B2`
`=A2^2`	`=SUM(A2:A100)`	`=PRODUCT(A2:A10)`	

8.2 Totals and averages

Formula	What it does
`=SUM(A2:A100)`	Total
`=AVERAGE(A2:A100)`	Mean
`=MEDIAN(A2:A100)`	Median
`=MIN(A2:A100)`	Smallest value
`=MAX(A2:A100)`	Largest value
`=COUNT(A2:A100)`	Counts numeric cells
`=COUNTA(A2:A100)`	Counts non-empty cells
`=COUNTBLANK(A2:A100)`	Counts empty cells

8.3 Rounding and numeric control

`=ROUND(A2;2)`	`=ROUNDUP(A2;2)`	`=ROUNDDOWN(A2;2)`	`=INT(A2)`
`=TRUNC(A2)`	`=ABS(A2)`	`=MOD(A2;3)`	`=SIGN(A2)`
`=SQRT(A2)`	`=POWER(A2;2)`	`=EXP(A2)`	`=LN(A2)`
`=LOG10(A2)`			

8.4 Logical functions

```
=IF(A2>=100;"Pass";"Review")
=AND(A2>0;B2>0)
=OR(A2="A";A2="B")
=NOT(A2="Closed")
=IF(AND(A2>=100;B2="North");"Priority";"Standard")
=IF(OR(A2="Urgent";B2>5000);"Review";"Normal")
```

IFS:

```
=IFS(A2>=90;"High";A2>=70;"Medium";A2<70;"Low")
```

If IFS is unavailable in an older version, use nested IF:

```
=IF(A2>=90;"High";IF(A2>=70;"Medium";"Low"))
```

Error handling:

```
=IFERROR(A2/B2;0)
=IFNA(VLOOKUP(E2;$A$2:$D$100;2;0);"Not found")
```

IFERROR catches many errors; IFNA catches the not-available lookup error when supported.

8.5 Conditional totals

SUMIF (one condition):

```
=SUMIF(A2:A100;"North";C2:C100)
```

SUMIFS (multiple conditions):

```
=SUMIFS(C2:C100;A2:A100;"North";B2:B100;"Retail")
```

SUMIFS with numerical threshold:

```
=SUMIFS(C2:C100;B2:B100;">100")
```

SUMIFS with date interval:

```
=SUMIFS(C2:C100;A2:A100;">="&DATE(2026;1;1);A2:A100;"<"&DATE(2026;2;1))
```

8.6 Conditional counts

```
=COUNTIF(A2:A100;"North")
=COUNTIF(C2:C100;">100")
=COUNTIFS(A2:A100;"North";B2:B100;"Retail")
=COUNTIFS(A2:A100;"North";C2:C100;">100")
```

8.7 Conditional averages and extrema

```
=AVERAGEIF(A2:A100;"North";C2:C100)
=AVERAGEIFS(C2:C100;A2:A100;"North";B2:B100;"Retail")
=MAXIFS(C2:C100;A2:A100;"North")
=MINIFS(C2:C100;A2:A100;"North")
```

MAXIFS, MINIFS, and AVERAGEIFS availability can depend on version. Check the Function Wizard or official help. If unavailable, use a Pivot Table, filter the data, or use an array formula only after checking your version's syntax.

8.8 Text and pattern criteria

COUNTIF with wildcard:

```
=COUNTIF(A2:A100;"AB*")
```

Counts cells beginning with `AB`. In wildcard mode, `?` matches one character and `*` matches any number of characters. Whether criteria use wildcards or regular expressions depends on Calc's calculation settings. For portability with Excel, wildcards are generally safer than Calc-specific regex criteria.

8.9 Useful text functions

```
=TRIM(A2)                    =CLEAN(A2)                   =UPPER(A2)
=LOWER(A2)                   =PROPER(A2)                  =LEFT(A2;4)
=RIGHT(A2;3)                 =MID(A2;2;5)                 =LEN(A2)
=FIND("-";A2)                =SEARCH("code";A2)           =SUBSTITUTE(A2;"old";"new")
=REPLACE(A2;3;2;"XY")        =CONCATENATE(A2;" ";B2)      =A2&" "&B2
=TEXT(A2;"0.00")             =VALUE(A2)
```

8.10 Date and time functions

```
=TODAY()          =NOW()           =DATE(2026;11;8)     =YEAR(A2)
=MONTH(A2)        =DAY(A2)         =HOUR(A2)            =MINUTE(A2)
=SECOND(A2)       =WEEKDAY(A2)     =EOMONTH(A2;0)       =DATEVALUE(A2)
=TIME(12;30;0)    =TIMEVALUE(A2)
```

A date/time is stored as a serial number. The integer part represents days; the fractional part represents time of day. Format the cell to display the desired date/time.

8.11 Lookup functions

```
=VLOOKUP(E2;$A$2:$D$100;2;0)
=HLOOKUP(E2;A1:Z4;3;0)
=INDEX(B2:B100;MATCH(E2;A2:A100;0))
=LOOKUP(E2;A2:A100;B2:B100)
=XLOOKUP(E2;A2:A100;B2:B100;"Not found")  [if supported by installed version]
```

For exact matching in VLOOKUP, use `0` or `FALSE` as the last argument. VLOOKUP searches the first column of its selected range and returns a column to the right. INDEX/MATCH can retrieve a value from a separate result range and

is more flexible.

8.12 Statistical functions

Formula	What it does
`=AVERAGE(A2:A100)`	Mean
`=MEDIAN(A2:A100)`	Median
`=MODE(A2:A100)`	Most frequent value
`=STDEV(A2:A100)`	Sample standard deviation in Calc
`=STDEV.P(A2:A100)`	Population standard deviation
`=VAR(A2:A100)`	Sample variance in Calc
`=VAR.P(A2:A100)`	Population variance
`=CORREL(A2:A100;B2:B100)`	Correlation
`=COVAR(A2:A100;B2:B100)`	Covariance
`=SLOPE(B2:B100;A2:A100)`	Regression slope
`=INTERCEPT(B2:B100;A2:A100)`	Regression intercept
`=RSQ(B2:B100;A2:A100)`	R-squared
`=STEYX(B2:B100;A2:A100)`	Standard error of the estimate
`=FORECAST.LINEAR(x;known_y;known_x)`	Predicted y [if supported by installed version]

Important Check the function's definition in Function Wizard. Excel distinguishes `STDEV.S` from `STDEV.P`. Calc's `STDEV` is the sample version; `STDEV.P` is population. Similar naming differences can exist in variance functions.

8.13 Trigonometry and angle functions

`=SIN(A2)`	`=COS(A2)`	`=TAN(A2)`	`=ASIN(A2)`
`=ACOS(A2)`	`=ATAN(A2)`	`=ATAN2(y;x)`	`=RADIANS(A2)`
`=DEGREES(A2)`	`=PI()`		

For `ATAN2`, check argument order in the Calc Function Wizard. Most spreadsheet trigonometric functions expect radians, not degrees. If A2 contains degrees:

```
=SIN(RADIANS(A2))
```

If a result is in radians and you want degrees: `=DEGREES(B2)`

8.14 Matrix and array-related functions

SUMPRODUCT multiplies corresponding entries and sums the products:

```
=SUMPRODUCT(A2:A100;B2:B100)
```

Example weighted total: `=SUMPRODUCT(B2:B10;C2:C10)`

Array formulas can behave differently between versions and compatibility modes. Use SUMPRODUCT or helper columns where possible because they are often easier to audit.

9 Lookup and Cross-Reference Methods

Example tables

Transactions sheet	**Product_List** sheet
A = Record ID B = Product Code C = Quantity	A = Product Code B = Description C = Unit Cost

9.1 VLOOKUP

In `Transactions.D2` (returns the description):

```
=VLOOKUP(B2;Product_List.$A$2:$C$100;2;0)
```

For unit cost:

```
=VLOOKUP(B2;Product_List.$A$2:$C$100;3;0)
```

Safe version:

```
=IFERROR(VLOOKUP(B2;Product_List.$A$2:$C$100;2;0);"Not found")
```

9.2 INDEX / MATCH

```
=INDEX(Product_List.$B$2:$B$100;MATCH(B2;Product_List.$A$2:$A$100;0))
```

How it works:

1. MATCH finds the row position of the code.
2. INDEX returns the value at that position from the description range.

9.3 Why a lookup returns #N/A

Check:

- Does the key exist?
- Are both keys the same data type (text versus number)?
- Are there leading/trailing spaces?
- Are leading zeros preserved?
- Is the lookup range correct?
- Did the lookup range move after copying the formula? Use `$` signs.
- Did you use exact match (`0`)?
- Are there duplicate keys in the lookup table?

10 Statistics, Regression, Covariance, and Uncertainty Calculations

This section covers common calculations in practical data analysis.

10.1 Mean

```
=AVERAGE(A2:A3505)
```

10.2 Sample variance

```
=VAR(A2:A3505)
```

10.3 Population variance

```
=VAR.P(A2:A3505)
```

10.4 Sample standard deviation

```
=STDEV(A2:A3505)
```

10.5 Covariance

```
=COVAR(A2:A3505;B2:B3505)
```

COVAR calculates **population** covariance in Calc. Verify whether your task requires sample or population covariance. If sample covariance is required and a dedicated function is not available in your version, calculate:

```
=SUMPRODUCT(A2:A3505-AVERAGE(A2:A3505);B2:B3505-AVERAGE(B2:B3505))/(COUNT(A2:A3505)-1)
```

Depending on version, array evaluation may be needed for the expression inside SUMPRODUCT. A safer helper-column method: in C2,

```
=(A2-AVERAGE($A$2:$A$3505))*(B2-AVERAGE($B$2:$B$3505))
```

Then sample covariance:

```
=SUM(C2:C3505)/(COUNT(A2:A3505)-1)
```

10.6 Correlation

```
=CORREL(A2:A3505;B2:B3505)
```

Range is from −1 to +1. Correlation measures linear association, not causation.

10.7 Linear regression

Formula	What it does
`=SLOPE(B2:B3505;A2:A3505)`	Slope
`=INTERCEPT(B2:B3505;A2:A3505)`	Intercept
`=$F$1*D2+$F$2`	Predicted y for x in D2, where F1 contains the slope and F2 the intercept

Equation: `y = a*x + b`

> **Warning** Do not reverse the arguments: `SLOPE(known_y;known_x)`.

10.8 Deming regression and errors in both variables

Ordinary least-squares regression assumes the x-values are known without measurement error. If both x and y have significant uncertainty, a different model such as Deming regression may be required. Do not automatically use SLOPE as a substitute for Deming regression. If a task gives a specified ratio of error variances, use the exact formula supplied by the task or an approved numerical method. Keep the input uncertainty columns and document the ratio assumption.

10.9 Uncertainty propagation by partial derivatives

For independent input quantities x and y and output f(x,y):

```
sigma_f = SQRT((df/dx*sigma_x)^2 + (df/dy*sigma_y)^2)
```

Function	Uncertainty
`f = x + y`	`sigma_f = SQRT(sigma_x^2 + sigma_y^2)`
`f = x - y`	`sigma_f = SQRT(sigma_x^2 + sigma_y^2)`
`f = x*y`	`sigma_f/ABS(f) = SQRT((sigma_x/x)^2 + (sigma_y/y)^2)  (relative uncertainty)`
`f = x/y`	`sigma_f/ABS(f) = SQRT((sigma_x/x)^2 + (sigma_y/y)^2)  (relative uncertainty)`

These formulas assume independent errors and small uncertainties. If errors are correlated, covariance terms may be required.

10.10 Filter by uncertainty threshold

Suppose a dataset has 6,793 records and an uncertainty field for each of several measurements. Keep only records where every required uncertainty is `<= 0.05`.

1. **Data > AutoFilter.**
2. Set each uncertainty column to `<= 0.05`.
3. Apply all filters together. The conditions across columns act as AND.
4. Copy visible qualifying rows to a new sheet named `filtered_data`.
5. Check the remaining count.

In the worked example from an earlier practical workflow, the official filter result was **3,504 records from 6,793**. Treat those counts as a validation target only when using that same source dataset and same criteria. Do not use a synthetic 250-row sample as a substitute for a required full dataset.

10.11 Column / range discipline

If a header is in row 1 and there are 3,504 retained records, the data occupy rows 2 through 3,505. A range including the header is row 1 through 3,505.

Example	Meaning
`$M$1:$M$3505  $O$1:$O$3505`	Includes the header
`$M$2:$M$3505`	Excludes the text header (normal for statistics)

Always check whether your formula range accidentally includes a header or omits the final data row.

10.12 Useful validation checks

- `COUNT` of numeric rows.
- `COUNTBLANK` of required fields.
- `MIN` and `MAX` of each measurement.
- Check no uncertainty threshold is violated.
- Check that calculated columns contain no unexpected errors.
- Compare formulas against a few hand calculations.

11 Trigonometry and Angle Calculations

11.1 Degrees versus radians

Calc's SIN, COS, and TAN expect radians. If the angle in degrees is in A2:

```
=SIN(RADIANS(A2))
=COS(RADIANS(A2))
=TAN(RADIANS(A2))
```

11.2 Convert back to degrees

```
=DEGREES(ASIN(B2))
```

11.3 Two-argument inverse tangent

ATAN2 is useful when the sign and quadrant of x and y matter. Check the Function Wizard for the order of arguments in your Calc version. Do not assume all spreadsheet programs present the arguments in the same order. Test with known points:

Point	Expected angle
x positive, y zero	0 degrees
x zero, y positive	90 degrees
x negative, y zero	180 degrees or −180 degrees, depending on convention
x zero, y negative	−90 degrees

11.4 Normalize an angle

Formula	What it does
`=MOD(A2;360)`	Common 0-to-360-degree normalization
`=MOD(A2+180;360)-180`	-180-to-180-degree normalization

11.5 Difference between angles

For a signed smallest difference in degrees:

```
=MOD(B2-A2+180;360)-180
```

Test edge cases around 0 and 360 degrees.

11.6 Periodic calculations

For a generic repeating process: `phase = 2*PI()*time/period`. Then:

```
=SIN(2*PI()*A2/$B$1)
=COS(2*PI()*A2/$B$1)
```

Use these to build smooth repeating curves and compare two processes with different periods or phase offsets.

12 Dates, Times, and Continuous Time Values

12.1 Date serials

Calc stores dates as serial numbers. Subtracting two dates gives elapsed days.

Formula	What it does
`=B2-A2`	Elapsed days
`=(B2-A2)*24`	Elapsed hours
`=(B2-A2)*1440`	Elapsed minutes

12.2 Date/time construction

```
=DATE(2026;11;8)+TIME(14;30;0)
```

12.3 Extract components

```
=YEAR(A2)    =MONTH(A2)    =DAY(A2)
=HOUR(A2)    =MINUTE(A2)   =SECOND(A2)
```

12.4 Continuous day count from a chosen epoch

If a task requires a continuous day count, define the epoch explicitly. For an epoch stored in E1:

```
=A2-$E$1
```

This avoids hard-coding an unexplained offset. If the task specifies a particular epoch or time standard, follow that specification exactly.

12.5 Time-zone and daylight-saving caution

A spreadsheet date/time usually does not carry a full time-zone database. Store the original time zone or offset in a separate field when records come from different regions. Do not silently mix local times and UTC values.

13 Graphs: Complete Tutorial from Data to Polished Chart

Charts are often where a correct analysis can be made to look wrong. The most important choice is selecting the correct chart type and defining the X and Y ranges explicitly.

13.1 Choose the correct chart

Chart type	Use it to...
Column	Compare categories, such as totals by department.
Bar	Compare categories when labels are long.
Line	Show a sequence over equally spaced categories, such as monthly totals.
XY Scatter	Show the relationship between two numerical variables. Use this when the x-values are real numerical values, not just labels.
Pie	Show a small number of parts of one whole; avoid many slices.
Area	Show totals or cumulative quantities over time.
Combination	Compare different measures, sometimes with different scales, if the chart remains readable.

> **Warning** For two numerical variables, use **XY Scatter**, not a Line chart.

13.2 Prepare graph data correctly

A clean scatter table:

X_Value	Series_A	Series_B
0	2.1	1.8
1	2.8	2.4
2	3.0	2.9
3	4.2	3.7

Situation	Layout
One series	X values in column A. Y values in column B.
Two series sharing the same x-values	X values in A. Series A y-values in B. Series B y-values in C.
Two series with different x-values	Keep each series' x and y columns together, for example A:B and D:E. Add each series separately in the chart's Data Series page.

13.3 Create a chart

1. Select the data including headers.
2. **Insert > Chart.**
3. Chart Wizard opens.
4. Choose chart type.
5. For numerical x-y data, choose **XY (Scatter)**.
6. Choose points only or points and lines, depending on the task.
7. Click Next.
8. Inspect Data Range.
9. Choose **Data series in columns** when each series is a column.
10. Set **First row as label** and **First column as label** if appropriate.
11. Click Next to inspect Data Series.
12. Verify each series name, X-values range, and Y-values range.
13. Click Next to add chart elements.
14. Add title, subtitle if needed, X-axis title, Y-axis title.
15. Click Finish.

13.4 The most important chart dialog: Data Series

If the graph is wrong, inspect:

- **Series list:** are all required series present?
- **Name:** does the series point to the header cell?
- **X-values:** does it point to the intended numeric x column?
- **Y-values:** does it point to the intended numeric y column?
- **Categories:** relevant mainly for category charts.
- **Data range:** are headers included correctly?

> **Tip** For a scatter plot, each series must have its own X-values and Y-values ranges.

13.5 Edit an existing chart

1. Click the chart once to select the chart object.
2. Double-click the chart to enter chart-edit mode.
3. Right-click the element you want to change.
4. Choose the relevant Format command.
5. Click outside the chart to leave chart-edit mode.

USEFUL EDITS

- Right-click chart background > **Format Chart Area**.
- Right-click plot area > **Format Plot Area**.
- Right-click axis > **Format Axis**.
- Right-click data series > **Format Data Series**.
- Right-click title > **Format Title**.
- Use **Insert > Titles** while in chart-edit mode if available.

13.6 Change the chart type after creation

Double-click chart. Right-click chart area or use **Format > Chart Type**, depending on version. Choose the new type. If switching to XY Scatter, recheck every series' X-values. Do not assume Calc will infer the correct x ranges.

13.7 Change dot/marker size

1. Double-click chart.
2. Click the data series once to select the series.
3. Right-click > **Format Data Series**.
4. Find the Line/Point or Data Points settings; the exact tab depends on chart type/version.
5. Select a marker symbol and adjust its size.
6. Apply.

If the option is not visible, check that you selected the data series rather than the plot area or chart frame.

13.8 Change line thickness and marker style

Right-click series > **Format Data Series**. Use Line settings for color, width, and style. Use Point/Marker settings for symbol and size. For a points-only scatter chart, set line style to **None** if a connecting line appears.

13.9 Set axis minimum, maximum, and interval

1. Double-click chart.
2. Right-click the horizontal axis > **Format Axis**.
3. Open the **Scale** tab.
4. Disable **Automatic** for Minimum and Maximum if manual bounds are required.
5. Enter the desired minimum and maximum.
6. Set major interval if needed.
7. Repeat for vertical axis.

Do not set an axis minimum to zero automatically for every chart. For scatter plots, use a range that shows the data honestly and clearly.

13.10 Change number formatting on axes

Right-click axis > **Format Axis > Numbers**. Choose Number, Percentage, Currency, Date, or Scientific. Set decimal places and thousands separators.

13.11 Add a trendline

1. Double-click chart.
2. Select the data series.
3. Right-click > **Insert Trend Line**.
4. Choose Linear or another justified model.
5. Enable equation and R-squared display if the version offers those options.
6. Explain what the fit means; do not treat R-squared as proof of causation.

13.12 Add error bars

Error bars are available for suitable chart types and series.

1. Double-click chart.
2. Select the series.
3. Use **Insert > Y Error Bars** or the series formatting options, depending on version.
4. Choose fixed value, percentage, standard deviation, or cell-range-based values if available.
5. If custom cell ranges are supported, point to the positive and negative uncertainty columns.
6. Check that the error bars use the correct units and scale.

If your version does not expose custom error-bar ranges in the interface, consult the version-specific chart guide and use the supported chart option rather than inventing a range syntax.

13.13 Add labels

Use **Insert > Data Labels** while the series is selected, or right-click the series > **Insert Data Labels**. Use labels only when they help interpretation. Too many labels make a chart unreadable.

13.14 Legend

Use **Insert > Legend** or the Chart Wizard elements page. Rename series by changing the header cell or the series Name reference.

13.15 Make a professional chart

- Descriptive title.
- Units on axes.
- Consistent font.
- Readable tick labels.
- Minimal gridlines.
- No unnecessary 3D.
- Distinguishable series markers.
- Legend only when needed.
- Sensible axis bounds.
- No unexplained abbreviations.
- Source and time period in a subtitle or note when appropriate.

13.16 Put multiple datasets on the same graph

Case A: same X values

Use one shared X column and separate Y columns. Create XY Scatter. In Data Series, define Series A and Series B with the same X range and different Y ranges.

Case B: different X values

Create the chart from one series first. Double-click chart > right-click > **Data Ranges** or use the Data Series page. Add a second series. Set its X-values to its own X column and Y-values to its own Y column. Do not force both datasets to share one X range if their x-values differ.

14 Graph Troubleshooting: Wrong Axes, Missing Series, Bad Scales

The horizontal axis shows 1, 2, 3, 4 instead of my X data

LIKELY CAUSE

- You chose Line instead of XY Scatter.
- The X-values range was not assigned.

FIX

1. Double-click chart.
2. Change type to XY (Scatter).
3. Open Data Series.
4. Select each series.
5. Assign the correct X-values range.
6. Assign the correct Y-values range.
7. Confirm that the X-values are numeric, not text.

Two datasets appear as separate charts, or one dataset is missing

FIX

- Use one chart object.
- Add both series in the Data Series page.
- Define X and Y ranges separately for each series.
- Check that the second series is not accidentally assigned as categories.

The graph uses the wrong x-axis values

CHECK IN THIS ORDER

1. Chart type is XY Scatter.
2. Data Range page says data are in columns/rows correctly.
3. First row/column label options are correct.
4. Data Series page X-values point to the intended cells.
5. Y-values point to the intended cells.
6. X column contains numbers, not text.
7. The X and Y ranges have the same number of rows.
8. Header row is excluded from numerical ranges.

Dates are plotted as category labels, not proportional time

Use XY Scatter if spacing between dates matters numerically. Use Line if you intentionally want equally spaced categories.

Points are connected unexpectedly

Select the series > **Format Data Series > Line > None**, or choose Points Only in the chart subtype.

Markers are too large

Format Data Series > Point/Marker settings > reduce size.

A chart's x-axis is sorted alphabetically

Your chart may be using a category axis. Use XY Scatter for numeric x values. If categories are dates or month names, sort the source table chronologically before charting.

A series has a different scale

Do not immediately put it on a secondary axis. First ask whether comparing the scales is meaningful. If a secondary axis is necessary, use the series/axis options available in your Calc version and label both axes clearly.

The chart does not update after data changes

- Check that the chart data range includes the new rows.
- Right-click chart > **Data Ranges** and expand the source range.
- If the chart is based on a Pivot Table, refresh the Pivot Table first.

Axis numbers show too many decimals

Right-click axis > **Format Axis > Numbers** > set decimal places.

Chart wizard buttons are disabled

The data range syntax may be invalid. Check sheet name, separators, range boundaries, and quotation marks.

15 Pivot Tables / DataPilot

A Pivot Table summarizes many rows without writing a separate formula for every category.

15.1 Prepare data

- One header row.
- No blank header cells.
- No merged cells.
- No blank rows inside the table.
- Numeric columns must contain numeric values.
- Date column must contain real dates.

15.2 Create

1. Select a cell inside the dataset or select the full range.
2. **Data > Pivot Table > Insert or Edit** (menu wording can vary).
3. Choose **Current Selection**.
4. In the layout dialog, drag fields into **Row Fields**, **Column Fields**, **Data Fields**, and **Filter Fields**.
5. Double-click a field in Data Fields to choose Sum, Count, Average, Max, Min, etc.
6. Choose output location, often New Sheet.
7. Click OK.

15.3 Example layout

Rows	Region
Columns	Category
Data	Sum of Revenue
Filter	Year

15.4 Common aggregation mistakes

- Revenue should usually be **Sum**, not Count.
- Record ID should usually be **Count**, not Sum.
- A numeric ID may be summed accidentally; change its aggregation.
- If a number column is imported as text, Sum may behave unexpectedly.

15.5 Group dates

Grouping capabilities can differ by version and source. A robust approach is to add helper columns:

Formula	Helper column
`=TEXT(A2;"YYYY-MM")`	Month
`="Q"&ROUNDUP(MONTH(A2)/3;0)`	Quarter
`=YEAR(A2)`	Year

Then use those fields in the Pivot Table.

15.6 Filter

Drag a field to Filter Fields or use the filter dropdown in the resulting Pivot Table.

15.7 Refresh

Click inside the Pivot Table: **Data > Pivot Table > Refresh**. If new rows are outside the original source range, edit the source range first.

15.8 Edit

Click inside the Pivot Table: **Data > Pivot Table > Insert or Edit**. Adjust layout or source range.

15.9 Pivot Table troubleshooting

Symptom	Fix
Blank field names	Fix the source headers.
Wrong total	Check Sum versus Count.
New rows absent	Expand the source range and refresh.
Dates not grouping	Confirm they are real dates, not text.
Duplicate-looking categories	Clean spaces and inconsistent capitalization first.

16 Analysis Tools, Goal Seek, Scenarios, and Solver

16.1 Goal Seek

Use when one input must be adjusted to make a formula equal a target.

Cell	Contents
B2	Revenue (starting guess)
B3	Fixed cost
B4	Variable cost rate
B5	Profit `=B2-B3-B2*B4`

Tools > Goal Seek

Formula Cell	B5
Target Value	Desired profit
Variable Cell	B2

Calc changes B2 until B5 reaches the target.

16.2 Scenarios

Use when you want to compare several sets of assumptions.

1. Select the changing input cells.
2. **Tools > Scenarios.**
3. Name the scenario.
4. Change assumptions and create additional scenarios.
5. Switch between scenarios to compare outputs.

16.3 Solver

Solver handles optimization with constraints, if installed and available: **Tools > Solver**. Example: maximize profit by choosing quantities subject to budget and capacity constraints. Define:

- Objective cell.
- Maximize / minimize / target.
- Variable cells.
- Constraints.

> **Do not confuse** Goal Seek = one variable, one target. Solver = multiple variables and constraints.

16.4 Data > Statistics

Depending on version, Calc provides tools such as descriptive statistics, correlation, covariance, regression, sampling, and other analysis procedures. Use the **Data > Statistics** menu and inspect the available tool list in your version. For regression, identify the X and Y input ranges carefully and choose output location.

17 Spreadsheet Automation and Macros

17.1 Macro recording

Macro recording may need enabling: **Tools > Options > LibreOffice > Advanced** > Enable macro recording (limited), if available.

1. **Tools > Macros > Record Macro.**
2. Perform repetitive formatting/actions.
3. Stop recording.
4. Save with a meaningful name.

Run: Tools > Macros > Organize Macros > Basic. Select the macro and Run.

17.2 What macros can and cannot do

- The recorder is limited.
- Some actions are not recorded.
- Relative and absolute cell references matter.
- Excel VBA is not the same as LibreOffice Basic.
- UNO API provides deeper control.
- Macro security matters; run only trusted macros.

17.3 Safe automation habits

- Test on a copy.
- Use clear macro names.
- Keep a manual fallback.
- Do not store passwords in macros.
- Do not enable all macros globally just to run one file.
- Save a backup before running a macro that edits many cells.

18 Practical Workflow for a Timed Data-Analysis Task

Use this sequence. It reduces mistakes and avoids wasting time.

Phase 1: Inspect

1. Open the file.
2. Check row count and column count.
3. Identify headers and data types.
4. Check whether dates and numeric values are real numbers.
5. Identify missing values and duplicate keys.
6. Save a working copy.

Phase 2: Import and clean

1. Import CSV with correct delimiter and locale.
2. Preserve IDs as text if needed.
3. Standardize text.
4. Convert numbers and dates.
5. Filter by required criteria.
6. Remove only genuine duplicates.
7. Create a clean output sheet.
8. Record the number of retained rows.

Phase 3: Calculate

1. Add derived columns.
2. Use absolute references for constants.
3. Use helper columns for complex calculations.
4. Fill formulas down.
5. Check first row, middle row, and final row manually.
6. Check for errors.

Phase 4: Analyze

1. Calculate count, mean, median, min, max, variance, standard deviation.
2. Calculate covariance/correlation if needed.
3. Fit a regression if required.
4. Use SUMIFS/COUNTIFS/AVERAGEIFS for grouped results.
5. Build Pivot Tables if useful.

Phase 5: Graph

1. Decide whether data are categories or numerical x-y pairs.
2. Choose chart type.
3. Define X and Y ranges explicitly.
4. Add every required series.
5. Set axis bounds and units.
6. Add titles and legend.
7. Check whether the graph supports the intended conclusion.

Phase 6: Validate and save

1. Check row counts.
2. Check formulas and units.
3. Recalculate.
4. Save ODS.
5. Export XLSX only if needed.
6. Reopen and inspect the delivered file.

19 Exact Troubleshooting Notes and Common Calc Errors

ERROR CODES AT A GLANCE

	Error	Meaning	What to check
19.1	`Err:509`	Missing operator.	Whether a multiplication sign, plus sign, or separator is missing.
19.2	`Err:508`	Unmatched pair.	Parentheses.
19.3	`Err:502`	Invalid argument.	SUMIFS ranges with different sizes; wrong argument type; invalid range; incorrect argument order.
19.4	`#NAME?`	Calc does not recognize a function or name.	Spelling; function availability in your version; named range exists; text values are in quotation marks; function language/name settings.
19.5	`#N/A`	Lookup did not find a match.	Exact match, spaces, data type, leading zeros, and lookup range.
19.6	`#VALUE!`	Wrong data type.	Multiplying text by a number; a date imported as text; a formula referencing a text label unexpectedly.
19.7	`#DIV/0!`	Denominator is zero or blank.	Use `=IFERROR(A2/B2;0)`, but do not hide a meaningful divide-by-zero problem without documenting it.

19.8 Formula displays as text

Possible causes: cell format is Text; formula begins with an apostrophe; missing equals sign. **Fix:**

1. **Format > Cells > Numbers > General.**
2. Re-enter the formula.
3. Check for a leading apostrophe.

19.9 Formula does not update

Check:

- Automatic calculation enabled.
- Formula references correct cells.
- Workbook is not in manual calculation mode.
- Press **F9** to recalculate.

19.10 Decimal comma versus decimal point

Check locale and import settings. Do not mix decimal conventions in one column.

19.11 A value looks rounded

Cell display format may hide decimals. Increase decimal places or inspect the formula bar. Displayed precision is not necessarily stored precision.

19.12 Sorting breaks rows

Undo immediately. Select the entire table before sorting, not a single field column.

19.13 Chart uses row numbers as X

Switch to XY Scatter and explicitly set X-values.

19.14 Chart series are wrong

Open Data Series and inspect Name, X-values, and Y-values for every series.

19.15 Formula separator confusion

Default Calc formulas usually use semicolons. If a copied formula has commas between arguments, adapt them. Do not change separators inside quoted strings.

19.16 Sheet-name syntax

Native Calc often uses `Sheet1.A1` rather than `Sheet1!A1`. Quote sheet names containing spaces:

```
='Clean Data'.A1
```

19.17 Data imported as text

Use Text to Columns or VALUE after checking locale and formatting.

19.18 Missing values treated as zero

A blank cell, an empty string, text `"NA"`, and numeric zero are different. Test using `ISBLANK`, `A2=""`, `COUNTBLANK`, and `ISNUMBER` as appropriate.

20 Excel-to-Calc Quick Reference

Task	Excel example	Calc example
Sum	`=SUM(A1:A10)`	`=SUM(A1:A10)`
IF	`=IF(A1>5,"Yes","No")`	`=IF(A1>5;"Yes";"No")`
AND	`=AND(A1>5,B1<10)`	`=AND(A1>5;B1<10)`
SUMIF	`=SUMIF(A:A,"North",B:B)`	`=SUMIF(A:A;"North";B:B)`
SUMIFS	`=SUMIFS(C:C,A:A,"North",B:B,"A")`	`=SUMIFS(C:C;A:A;"North";B:B;"A")`
COUNTIFS	`=COUNTIFS(A:A,"North",B:B,">5")`	`=COUNTIFS(A:A;"North";B:B;">5")`
VLOOKUP	`=VLOOKUP(E2,A:D,2,FALSE)`	`=VLOOKUP(E2;A:D;2;0)`
INDEX/MATCH	`=INDEX(B:B,MATCH(E2,A:A,0))`	`=INDEX(B:B;MATCH(E2;A:A;0))`
IFERROR	`=IFERROR(A1/B1,0)`	`=IFERROR(A1/B1;0)`
TRIM	`=TRIM(A2)`	`=TRIM(A2)`
CONCATENATE	`=CONCATENATE(A2," ",B2)`	`=CONCATENATE(A2;" ";B2)`
SUBSTITUTE	`=SUBSTITUTE(A2,"x","y")`	`=SUBSTITUTE(A2;"x";"y")`
ROUND	`=ROUND(A2,2)`	`=ROUND(A2;2)`
SIN	`=SIN(RADIANS(A2))`	`=SIN(RADIANS(A2))`
CORREL	`=CORREL(A2:A10,B2:B10)`	`=CORREL(A2:A10;B2:B10)`

IMPORTANT COMPATIBILITY NOTES

- Excel and Calc function support is not identical.
- `XLOOKUP` and newer dynamic-array functions depend on the Calc version.
- Excel macros do not automatically become LibreOffice macros.
- Formula separators can be changed by locale/settings.
- Some chart features and external connections may not round-trip perfectly.

21 How to Use Documentation Without AI

Documentation is not just something to read after you get stuck. It is a tool you can use systematically during practice.

21.1 First source: built-in Help

Press **F1** or choose **Help > LibreOffice Help**. Search for the exact task, not a vague phrase. Examples:

```
"Pivot Table"           "XY scatter chart"      "Goal Seek"             "REGEX function"
"Text to Columns"       "absolute reference"    "error 502"
```

Use Help when you need:

- A menu path.
- A function's argument order.
- A definition of a dialog option.

- A version-specific explanation.
- An explanation of an error code.

21.2 Function Wizard

Press **Ctrl+F2**. Workflow:

1. Search for the function name.
2. Read the function description.
3. Inspect every argument.
4. Identify which arguments are required.
5. Enter a tiny test example.
6. Compare the result with a hand calculation.
7. Only then apply it to thousands of rows.

This is especially useful for functions whose argument order is easy to confuse, such as SLOPE, INDEX, MATCH, VLOOKUP, ATAN2, and statistical functions.

21.3 Official Calc Guide

Use the official Calc Guide: https://books.libreoffice.org/en/CG262/CG262.html

It includes chapters on:

- Entering and formatting data.
- Formulas and functions.
- Charts and graphs.
- Pivot Tables.
- Data analysis.
- Linking data.
- Macros.
- Calc as a database.
- Keyboard shortcuts.
- Error codes.

If your installed version is older, use the matching guide version rather than assuming every screenshot or feature is identical.

21.4 Official Help website

https://help.libreoffice.org/

Search for a function by exact name. For example, SUMIFS:

help.libreoffice.org/latest/en-US/text/scalc/01/func_sumifs.html

The function reference normally provides:

- Syntax.
- Argument descriptions.
- Examples.
- Availability information.
- Notes about special criteria behavior.

21.5 The five-question method for a function

Whenever you look up a function, answer these five questions:

1. What does it return?
2. What is the exact syntax?
3. What does each argument mean?
4. What happens with blank cells, text, or errors?
5. Is it available in my installed version?

Then create a tiny test table with 3–5 rows and verify it.

21.6 Search the manual effectively

Use specific searches:

- "LibreOffice Calc SLOPE function arguments"
- "LibreOffice Calc chart data series X values"
- "LibreOffice Calc remove duplicates"
- "LibreOffice Calc regular expressions formulas"
- "LibreOffice Calc Pivot Table refresh"
- "LibreOffice Calc CSV import decimal separator"

PREFER, IN THIS ORDER

1. Official LibreOffice Help.
2. Official LibreOffice Guide.
3. Ask LibreOffice community answers.
4. Reputable tutorials that show the same version you use.

21.7 Ask the community effectively

Useful community site: https://ask.libreoffice.org/

Before posting:

- State your LibreOffice version (Help > About).
- State your operating system.
- Paste a minimal example dataset with fake values.
- Paste the exact formula.
- Paste the exact error.
- Explain what result you expected.
- Explain what happened instead.
- Include the menu path you tried.
- Do not upload private customer or company data.

> **A good question** "In LibreOffice Calc 26.x on Windows, I have numeric X values in A2:A101 and Y values in B2:B101. My XY Scatter chart shows 1–100 on the horizontal axis. How do I assign A2:A101 as the X-values for the series?"
> This is much better than: "My chart is broken."

21.8 Build your own offline reference folder

Create a folder with:

- Calc Guide PDF.
- Function reference notes.
- A small formula test workbook.
- A chart test workbook.
- A list of errors and their fixes.
- A text file of useful menu paths.

Download official documentation from: documentation.libreoffice.org/en/english-documentation/

If you may be offline, install the LibreOffice offline Help component if it is available for your operating system.

21.9 Documentation during a timed practical task

Do not read an entire chapter while the clock is running. Use this sequence:

1. Identify the exact operation.
2. Search the exact command/function name.
3. Read syntax or menu steps.
4. Test on 3 rows.
5. Apply to full data.
6. Validate with a second method.
7. Record the useful solution in your own notes.

21.10 Keep a personal error log

Use a table:

Date	Problem	Cause	Fix	Verification
Today	X axis shows 1, 2, 3	Line chart selected	Changed to XY Scatter and assigned X range	X axis now matches source values

This turns every mistake into a reusable skill.

22 Practice Drills and Final Checklist

Drill 1: CSV import

- Import a comma-separated file.
- Preserve an ID with leading zeros.
- Verify numeric columns with ISNUMBER.
- Verify date columns with ISNUMBER.
- Save as ODS.

Drill 2: Cleaning

- Remove leading/trailing spaces.
- Standardize labels.
- Replace a character.
- Remove duplicates using a defined key.
- Flag missing values.
- Count valid records.

Drill 3: Formulas

- Calculate sum, mean, median, variance, and standard deviation.
- Use IF and nested IF.
- Use AND and OR.
- Use SUMIFS and COUNTIFS.
- Use VLOOKUP and INDEX/MATCH.
- Use IFERROR.

Drill 4: Statistics

- Calculate covariance and correlation.
- Calculate regression slope and intercept.
- Plot predicted values.
- Compare a few results manually.

Drill 5: Graphs

- Create a column chart.
- Create a line chart.
- Create an XY scatter chart.
- Add two separate series.
- Correct the X-values range.
- Change marker size.
- Set axis limits.
- Add titles and a trendline.

Drill 6: Reporting

- Create a Pivot Table.
- Summarize by category.
- Filter by period.
- Refresh after changing source data.
- Build a one-sheet dashboard.

Drill 7: What-if

- Use Goal Seek.
- Create three scenarios.
- Compare output values.
- Record assumptions.

Final pre-submission checklist

- ☐ Original data preserved.
- ☐ CSV delimiter and locale correct.
- ☐ IDs preserved as intended.
- ☐ Dates are real dates.
- ☐ Numeric columns are numeric.
- ☐ Missing values handled transparently.
- ☐ Duplicate definition is justified.
- ☐ Filter criteria applied correctly.
- ☐ Row count checked.
- ☐ Formula ranges include all records and exclude headers where required.
- ☐ Absolute references used where needed.
- ☐ Functions tested on a small sample.
- ☐ No unexplained formula errors.
- ☐ Statistical method matches the requested definition.
- ☐ Graph type matches the data.
- ☐ Scatter X-values explicitly assigned.
- ☐ Every required series appears.
- ☐ Axes, units, and labels are clear.
- ☐ Pivot Tables refreshed.
- ☐ Results independently checked.
- ☐ ODS master saved.
- ☐ Exported file reopened and inspected.

23 Official Documentation and Learning Links

1	**LibreOffice Calc Guide (current guide landing page)** https://books.libreoffice.org/en/CG262/CG262.html
2	**LibreOffice Help** https://help.libreoffice.org/
3	**Official documentation downloads** https://documentation.libreoffice.org/en/english-documentation/
4	**Ask LibreOffice community** https://ask.libreoffice.org/
5	**LibreOffice FAQ** https://wiki.documentfoundation.org/Faq
6	**Calc chart guide** https://books.libreoffice.org/en/CG262/CG26206-CreatingChartsAndGraphs.html
7	**Formula basics** https://help.libreoffice.org/latest/en-US/text/scalc/guide/formulas.html
8	**SUMIFS function reference** https://help.libreoffice.org/latest/en-US/text/scalc/01/func_sumifs.html
9	**MAXIFS function reference** https://help.libreoffice.org/latest/en-US/text/scalc/01/func_maxifs.html
10	**Wildcards in formulas** https://help.libreoffice.org/latest/en-US/text/scalc/guide/wildcards.html

Final Advice

Learn the workflow, not just isolated formulas:

IMPORT	INSPECT	CLEAN	FILTER	CALCULATE	ANALYZE	GRAPH	VALIDATE	SAVE

Remember When a result looks wrong, do not immediately rewrite the formula. Check the input data type, locale, selected range, formula separators, chart type, and source range first. Most spreadsheet mistakes come from one of those foundations.

Appendix A Notes Carried Forward from Your Earlier Spreadsheet Work

This appendix preserves useful spreadsheet-specific details from your earlier practical work while keeping the examples and labels general.

A.1 Large-table filtering checkpoint

Original record count	6,793
Required rule	Every required uncertainty field must be less than or equal to 0.05.
Expected retained records	3,504 (for that exact source and exact rule)
Output sheet name used	`filtered_data`
Final data row	3,505 (if row 1 contains headers and 3,504 records remain)

- A full column range including the header can be `$M$1:$M$3505` or `$O$1:$O$3505`.
- A calculation range that excludes the header should begin at row 2, for example `$M$2:$M$3505`.
- Do not accidentally include the header in AVERAGE, VAR, CORREL, SLOPE, or other numeric calculations.

A.2 Filtering several columns at once

1. Keep the original table unchanged.
2. Turn on **Data > AutoFilter**.
3. In each required uncertainty column, select the numeric condition `<= 0.05`.
4. Apply all required column filters together. The record must pass every condition.
5. Copy the visible qualifying records to a new sheet.
6. Count the resulting records.
7. Confirm that no retained row violates any of the conditions.
8. Keep the filter criteria documented beside the result.

A.3 Example calculation checkpoints from the earlier working sheet

Warning These values are historical checkpoints from one earlier calculation. They are not universal constants. Use them only when reproducing the same data and the same definitions.

Checkpoint	Approximate value
Metric X mean	`-0.08998`
Metric X variance	`0.010503`
Metric Y mean	`-0.37631`
Metric Y variance	`0.016562`
Covariance between the two metric columns	`-0.00194`
Previously calculated fitted-line slope	`-3.41231`

If your results differ, check these items before changing the formula:

1. Did you use the same filtered records?
2. Did you use the same two columns?
3. Did you exclude the header row?
4. Did you use sample variance or population variance?
5. Did you calculate covariance with the same definition?
6. Did SLOPE receive Y values first and X values second?
7. Did the ranges start and end on the same rows?
8. Did a formula include an extra blank or text cell?

A.4 Covariance using two columns

If the two data columns are K and L and data occupy rows 2 to 3505:

```
=COVAR(K2:K3505;L2:L3505)
```

Use the exact same row limits for both columns. Do not use `K1:L3505` if row 1 contains headers.

A.5 Regression slope

If K contains X and L contains Y:

```
=SLOPE(L2:L3505;K2:K3505)
```

The order is known_y first, known_x second. If the task defines the reverse relationship, reverse the ranges deliberately.

A.6 Variance and mean

```
=AVERAGE(K2:K3505)
=VAR(K2:K3505)
=AVERAGE(L2:L3505)
=VAR(L2:L3505)
```

Remember:

- `VAR` is sample variance in Calc.
- `VAR.P` is population variance.
- A sample/population mismatch can explain a small numerical difference.

A.7 Graph setup that previously caused trouble

For a scatter graph:

- Use **XY (Scatter)**, not Line.
- Set the intended numeric horizontal variable as X-values.
- Set the intended vertical variable as Y-values.
- If there are two series, add both to the same chart object.
- Do not let Calc guess the X-values if the graph is important; set them explicitly in Data Series.
- Reduce marker size in Format Data Series > Point/Marker settings.
- If the horizontal axis shows 1, 2, 3, … rather than your actual X data, inspect the chart type and X-values range first.

A.8 Two separate datasets on one chart

When two datasets have different horizontal values:

1. Make an XY Scatter chart using the first dataset.
2. Double-click the chart.
3. Open the Data Series page through the chart data-range controls.
4. Add the second series.
5. Set the second series Name.
6. Set its X-values to the second dataset's X column.
7. Set its Y-values to the second dataset's Y column.
8. Confirm both series appear in the same chart.
9. Check the legend and axis ranges.

A.9 When a graph uses the wrong horizontal axis

Do not start by editing the numbers in the data table. Check in this order:

1. XY Scatter chart type.
2. Data are arranged in columns or rows correctly.
3. Header row and header column options.
4. Series X-values.
5. Series Y-values.
6. Numeric data type of the X column.
7. Equal number of X and Y records.

A.10 Multiple IF construction through the Function Wizard

If nested IF is confusing:

1. Select the result cell.
2. Press `Ctrl+F2`.
3. Search for IF.
4. Enter the first condition.
5. Enter the result for TRUE.
6. In the FALSE argument, enter another IF function.
7. Continue nesting IF functions until all cases are covered.
8. Confirm the formula and test one example from every category.

Example:

```
=IF(A2>=90;"High";IF(A2>=70;"Medium";IF(A2>=50;"Basic";"Low")))
```

The Function Wizard helps you insert the function structure, but you still need to understand the decision order. Put the highest threshold first when conditions are nested from high to low.

Appendix B Quick Function Checklist for Practical Data Work

Import and check	ISNUMBER · ISTEXT · ISBLANK · COUNT · COUNTA · COUNTBLANK
Clean and transform text	TRIM · CLEAN · SUBSTITUTE · REPLACE · REGEX · LEFT · RIGHT · MID · LEN · FIND · SEARCH · UPPER · LOWER · PROPER · CONCATENATE · TEXT · VALUE
Logic and errors	IF · IFS · AND · OR · NOT · IFERROR · IFNA
Conditional analysis	SUMIF · SUMIFS · COUNTIF · COUNTIFS · AVERAGEIF · AVERAGEIFS · MAXIFS · MINIFS
Lookups	VLOOKUP · HLOOKUP · INDEX · MATCH · XLOOKUP (if supported)
Basic statistics	SUM · AVERAGE · MEDIAN · MODE · MIN · MAX · VAR · VAR.P · STDEV · STDEV.P · CORREL · COVAR · SLOPE · INTERCEPT · RSQ · STEYX · FORECAST.LINEAR (if supported)
Math and angles	ABS · ROUND · ROUNDUP · ROUNDDOWN · INT · TRUNC · MOD · SIGN · SQRT · POWER · EXP · LN · LOG10 · SIN · COS · TAN · ASIN · ACOS · ATAN · ATAN2 · RADIANS · DEGREES · PI
Dates and time	DATE · DATEVALUE · TIME · TIMEVALUE · TODAY · NOW · YEAR · MONTH · DAY · HOUR · MINUTE · SECOND · WEEKDAY · EOMONTH

Appendix C The 60-Second Debug Routine

When a formula, chart, or summary looks wrong, do not randomly change settings.

1. Check the source cell values. Are they really numbers, dates, or text?
2. Check the imported delimiter and locale.
3. Check formula argument separators.
4. Check parentheses and quotation marks.
5. Check whether ranges include the header.
6. Check that paired X and Y ranges have equal lengths.
7. Check whether absolute references are required.
8. Check whether a lookup key contains spaces or leading zeros.
9. Check whether the chart is XY Scatter when X is numerical.
10. Check whether the Pivot Table has been refreshed.
11. Recalculate with `F9`.
12. Test the formula on 3–5 rows and compare with a hand calculation.

END OF MANUAL